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1)

$$\int (x^2 - 1)^2 dx = \int x^4 - 2x^2 + 1 dx = \int x^4 dx + \int -2x^2 dx + \int 1 dx = \frac{x^5}{5} - \frac{2x^3}{3} + x + C$$

2)

$$\int \frac{1}{x+1} + \frac{2}{1+x^2} - \frac{2}{\sqrt{x^2-1}} dx = \int \frac{1}{x+1} + \int \frac{2}{1+x^2} + \int -\frac{2}{\sqrt{x^2-1}} = \lg(|x+1|) + 2 \arctan x - 2 \operatorname{arcsinh} x + C$$

3)

$$\int 2 \sin(3x) + 5 \cos(2x) dx = 2 \int \sin(3x) + 5 \int \cos(2x) = -\frac{2}{3} \cos 3x + \frac{5}{2} \sin(2x) + C$$

4)

$$\int \frac{x}{x^2-2} dx = \frac{1}{2} \int \frac{2x}{x^2-2} dx = \frac{1}{2} \ln(x^2-2) + C$$

5)

$$\begin{aligned} \int \sin^3 x &= \int \sin x \sin^2(x) = \int \sin x (1 - \cos^2 x) dx = \int \sin x (1 - y^2) \frac{dy}{-\sin x} = -\int (1 - y^2) dy \\ &= \int -1 + y^2 dy = \int -1 + \cos^2 x \quad \begin{matrix} y = \cos x \\ dy = -\sin x dx \\ \frac{dy}{-\sin x} = dx \end{matrix} \\ \int -1 dy + \int y^2 dy &= -y + \frac{y^3}{3} = -\cos x + \frac{\cos^3 x}{3} \end{aligned}$$

6)

$$\int x^2 \operatorname{arctan} x = \frac{x^3}{3} \operatorname{arctan} x - \int \frac{x^3}{3} \cdot \frac{1}{1+x^2} dx = \frac{x^3}{3} \operatorname{arctan} x + \frac{1}{3} \int \frac{x^3}{x^2+1} dx =$$

$$\begin{aligned} \frac{x^3}{3} \operatorname{arctan} x + \frac{1}{3} \int \frac{x^3}{x^2+1} &= \frac{x^3}{3} \operatorname{arctan} x + \frac{1}{6} \int \frac{x^2-1+1}{x^2+1} dx = \frac{x^3}{3} \operatorname{arctan} x + \frac{1}{6} \int \frac{y-1}{1} dy - \\ y &= x^2-1 \\ dy &= 2x dx \\ \frac{dy}{2x} &= dx \end{aligned}$$

$$= \frac{x^3}{3} \arctan x + \frac{1}{6} \int \frac{y-1}{y} = \frac{x^3}{3} \arctan x - \frac{1}{6} \int 1 - \frac{1}{y} = \frac{x^3}{3} \arctan x - \frac{1}{6} \left(\int 1 dy - \int \frac{1}{y} dy \right) = \frac{x^3}{3} \arctan x - \frac{1}{6} (y - \ln|y|)$$

$$\frac{x^3}{3} \arctan x - \frac{1}{6} (y - \ln|y|) = \frac{x^3}{3} \arctan x - \frac{1}{6} y + \ln y = \frac{x^3}{3} \arctan x - \frac{x^2+1}{6} + \ln(x^2+1) + C$$

$$7) \int \frac{x^3}{x^2+x-2} dx$$

$$\int x - 1 + \frac{3x-2}{x^2+x-2} dx$$

$$\int x - \int 1 + \int \frac{3x-2}{x^2+x-2} dx$$

$$\frac{x^2}{2} - x + \int \frac{4}{x-2} - \int \frac{1}{x-1}$$

$$\frac{x^2}{2} - x + 4 \lg|x-2| - \lg|x-1| + C$$

$$\frac{x^3}{x^2+x-2} = \frac{x^3}{x^2+x-2} \cdot \frac{x-1}{x-1} = \frac{-x^3 - x^2 + 2x}{-x^2 + 2x}$$

$$\frac{-x^3 - x^2 + 2x}{-x^2 + 2x} = \frac{-x^2 + x - 2}{3x - 2}$$

$$x^2+x-2 = (x+2)(x-1)$$

$$\frac{A}{(x-2)} + \frac{B}{x-1} = \frac{3x-2}{(x+2)(x-1)}$$

$$A(x-1) + B(x-2) = 3x-2$$

$$Ax - A + Bx - 2B = 3x - 2$$

$$x(A+B) - A - 2B = 3x - 2$$

$$\begin{cases} A+B=3 \\ -A-2B=-2 \end{cases}$$

$$\begin{cases} A=3-B \\ -3+B-2B=-2 \end{cases}$$

$$\begin{cases} A=3-B \\ -B=1 \end{cases}$$

$$\begin{cases} A=3-B \\ B=-1 \end{cases}$$

$$\begin{cases} A=4 \\ B=-1 \end{cases}$$

8)

$$\int \underset{\substack{\text{f} \\ \text{G}}}{x} \underset{\text{F}}{e^{-3x}} = -\underset{\text{G}}{x} \underset{\text{F}}{\frac{1}{3}} e^{-} - \int \frac{1}{3} e^{-3x} = -\frac{x}{3} e^{-3x} - \frac{1}{9} e^{-3x} + C$$

$$\int e^{-3x} dx = -\frac{e^y}{3} = -\frac{e^{-3x}}{3} = -\frac{1}{3} e^{-3x}$$

$$y = -3x$$

$$dy = -3 dx$$

$$-\frac{dy}{3} = dx$$

9)

$$\int \underset{\substack{\text{f} \\ \text{G}}}{x} \underset{\text{F}}{\lg^2 x} dx = \frac{x^2}{2} \underset{\text{G}}{\lg^2 x} - \int \frac{x^2}{2} \cdot \frac{2 \lg x}{x} = \frac{x^2}{2} \lg^2(x) - \int \frac{\cancel{2} x^{\cancel{2}}}{\cancel{2} x} \lg x = \frac{x^2}{2} \lg^2 x - \int x \lg x =$$

$$\frac{x^2}{2} \lg^2 x - \int \underset{\substack{\text{f} \\ \text{G}}}{x} \underset{\text{F}}{\lg x} = \frac{x^2}{2} \lg^2(x) - \left(\frac{x^2}{2} \lg x - \int \frac{x^2}{2} \cdot \frac{1}{x} \right) = \frac{x^2}{2} \lg^2(x) - \left(\frac{x^2}{2} \lg x - \frac{1}{2} \int x \right) = \frac{x^2}{2} \lg^2(x) - \left(\frac{x^2}{2} \lg x - \frac{x^2}{4} \right) + C$$

$$= \frac{x^2}{2} \lg^2(x) - \left(\frac{x^2}{2} \lg x - \frac{x^2}{4} \right) + C = \frac{x^2}{2} \lg^2(x) - \frac{x^2}{2} \lg x + \frac{x^2}{4} + C$$

10)

$$\int \underset{\text{F}}{(x-1)} \underset{\text{f}}{\cos(x+1)} dx = (x-1) \sin(x+1) - \int \sin(x+1) dx = (x-1) \sin x + \cos(x+1) + C = x \sin(x+1) - \sin(x+1) + \cos(x+1) + C$$

$$\int \cos(x+1) = \sin(y) = \sin(x+1)$$

$$y = x+1$$

$$dy = dx$$

$$11) \int \underset{f}{y} \cos \underset{f}{y} dy = y \sin y - \int \sin y = y \sin y + \cos y + C$$

12)

$$\int \tan(x) + 3 \cotan(x) dx = \int \tan(x) + 3 \int \cotan(x) dx = \int \frac{\sin(x)}{\cos(x)} dx + 3 \int \frac{\cos(x)}{\sin(x)} dx = \int \frac{\sin(x)}{\cos(x)} \cdot \frac{1}{-\sin(x)} dx + 3 \int \cotan(x) dx$$

$$- \int \frac{1}{y} dy + 3 \int \cotan(x) dx = -\ln|y| + 3 \int \frac{\cos(x)}{\sin(x)} dx \quad \begin{matrix} y = \cos(x) \\ dy = -\sin(x) dx \end{matrix}$$

$$- \ln|y| + 3 \int \frac{\cos(x)}{\sin(x)} \cdot \frac{1}{\cos(x)} dx =$$

$$\begin{matrix} z = \sin(x) \\ dz = \cos(x) dx \\ \frac{dz}{\cos(x)} = dx \end{matrix}$$

$$\begin{matrix} z = \sin(x) \\ dz = \cos(x) dx \\ \frac{dz}{\cos(x)} = dx \end{matrix}$$

$$- \ln|y| + 3 \int \frac{1}{z} dz = -\ln|y| + 3 \ln|z| + C = -\ln|\cos(x)| + 3 \ln|\sin(x)| + C$$

15)

$$\int \frac{2x+1}{x^2+6x+9} dx = \int \frac{2x+1}{(x+3)^2}$$

$$\int -\frac{5}{(x+3)^2} + \frac{2}{x+3} dx$$

$$-5 \int \frac{1}{(x+3)^2} + 2 \int \frac{1}{x+3}$$

$$-5 \int \frac{1}{(x+3)^2} + 2 \ln|x+3|$$

$$\begin{matrix} y = x+3 \\ dy = dx \end{matrix}$$

$$y^{-2} = \frac{y^{-2+1}}{-2+1} = \frac{y^{-1}}{-1} = -\frac{1}{y}$$

$$-5 \int \frac{1}{y^2} + 2 \ln|x+3| = \frac{5}{y} + 2 \ln|x+3| + C = \frac{5}{(x+3)} + 2 \ln|x+3| + C$$

$$\frac{\frac{A}{(x+3)^2} + \frac{B}{x+3}}{A + \frac{Bx+3B}{(x+3)^2}} = \frac{2x+1}{(x+3)^2}$$

$$A + Bx + 3B = 2x + 1$$

$$\begin{cases} A + 3B = 1 \\ B = 2 \end{cases} \quad \begin{cases} A + 6 = 1 \\ B = 2 \end{cases} \quad \begin{cases} A = -5 \\ B = 2 \end{cases}$$

16)

$$\int \sin(3x) \cos(4x) dx = \int \frac{1}{2} [\sin(3x-4x) + \sin(3x+4x)] = \frac{1}{2} \int \sin(-x) + \sin(7x) = \frac{1}{2} \left(\int \sin(-x) + \sin(7x) \right)$$

$$\frac{1}{2} \left(\int \sin(-x) + \sin(7x) \right) = \frac{1}{2} \left(\cos x - \frac{1}{7} \cos(7x) \right) = \frac{1}{2} \cos x - \frac{1}{14} \cos(7x) + C$$

$$y = -x$$

$$dy = -dx$$

$$-dy = dx$$

13)

$$\int \frac{x}{(4-3x^2)^5} = \int \frac{x}{(4-3x^2)^5} \cdot -\frac{1}{6x} dx = -\frac{1}{6} \int \frac{1}{y^5} dy = \frac{1}{6} \cdot -\frac{1}{4y^4} = -\frac{1}{24y^4} = -\frac{1}{24(4-3x^2)^4} = -(24(4-3x^2)^4)^{-1} + C$$

$$y = 4-3x^2$$

$$dy = -6x dx$$

$$-\frac{dy}{6x}$$

$$\frac{1}{y^5} = y^{-5} = \frac{y^{-5+1}}{-5+1} = \frac{y^{-4}}{-4} = -\frac{1}{4y^4}$$

14)

Ex 2

$$1) \int_1^2 5x^2 + \sqrt{x} - \frac{1}{x} + 4 \, dx = \left[\frac{5}{3} x^3 + \frac{2}{3} x \sqrt{x} - \ln|x| + 4x \right]_1^2 = \left(5 \cdot \frac{8}{3} + \frac{2}{3} \cdot 2\sqrt{2} - \ln 2 + 8 \right) - \left(5 \cdot \frac{1}{3} + \frac{2}{3} - \ln 1 + 4 \right)$$

$$\int x^2 = \frac{x^3}{3}$$

$$\int x^{1/2} = \frac{x^{1/2+1}}{\frac{1}{2}+1} = \frac{x^{3/2}}{\frac{3}{2}} = \frac{2}{3} x \sqrt{x}$$

$$\int -1/x = -\ln|x|$$

$$\int 4 = 4x$$

$$\frac{40}{3} + \frac{4\sqrt{2}}{3} - \ln 2 + 8 - \frac{5}{3} - \frac{2}{3} - 4$$

$$\frac{33}{3} + \frac{4\sqrt{2}}{3} - \ln 2 + 4 = 11 + \frac{4\sqrt{2}}{3} - \ln 2 + 4 =$$

$$15 + \frac{4\sqrt{2}}{3} - \ln 2$$

$$2) \int_0^1 \frac{8x^3}{x^4+1} = \left[2 \ln(x^4+1) \right]_0^1 = (2 \ln(2)) - (2 \ln(1)) = 2 \ln 2 = \ln 4$$

$$\int \frac{8x^3}{x^4+1} = \int \frac{8x^3}{x^4+1} \cdot \frac{1}{4x^3} = \int \frac{2}{y} \, dy = 2 \int \frac{1}{y} \, dy = 2 \ln y = 2 \ln(x^4+1)$$

$$y = x^4 + 1$$

$$dy = 4x^3 \, dx$$

$$\frac{dy}{4x^3} = dx$$

$$3) \int_0^4 \frac{1}{2\sqrt{x}+1} \left[\sqrt{4} - \frac{1}{2} \ln |2\sqrt{x}+1| \right]_0^4 = 2 - \frac{1}{2} \ln 5$$

$$\int \frac{1}{2\sqrt{x}+1} = \int \frac{2y}{2\sqrt{y}+1} = \int 1 - \frac{1}{2\sqrt{y}+1} = y - \frac{1}{2} \int \frac{1}{z} = y - \frac{1}{2} \ln |z| = \sqrt{x} - \frac{1}{2} \ln |2\sqrt{x}+1|$$

$z = 2\sqrt{y}+1$ $dz/dx = 2$

$$y = \sqrt{x}$$

$$\frac{dy}{dx} = \frac{1}{2\sqrt{x}} \quad \sqrt{x} = \frac{1}{2} 2\sqrt{x}$$

$$\frac{dz}{dx} = 2 \quad \frac{dz}{2} = dx$$

$$2\sqrt{x} = dz$$

4) ???

5) ???

6)

$$\int_0^1 (2y+1)^4 dy = \left[\frac{(2y+1)^5}{5} \right]_0^1 = \frac{3^5}{5} - \frac{1}{5} = \frac{242}{5} - \frac{1}{5} = \frac{241}{5}$$

$$\int (2y+1)^4 dy = \int \frac{(z)^4}{2} = \frac{1}{2} \int z^4 = \frac{1}{2} \cdot \frac{z^5}{5} = \frac{(2y+1)^5}{10}$$

$$z = 2y+1 \quad dz/dy = 2$$

$$dz = 2 dy$$

$$\frac{dz}{2} = dy$$

8/3

$$\int_0^1 3^t - 2^{3t+1} + 5e^{-7t} dt = \left(\frac{3^t}{\ln 3} - \frac{2^{3t+1}}{3 \ln 2} - \frac{5}{7} e^{-7t} \right) - \left(\frac{3^0}{\ln 3} - \frac{2^1}{3 \ln 2} - \frac{5}{7} \right) = \frac{4}{\ln 3} - \frac{16}{3 \ln 2} + \frac{2}{3 \ln 2} - \frac{5}{7} e^{-7} + \frac{5}{7}$$
$$\int 3^t - \int 2^{3t+1} + \int 5e^{-7t} = \frac{3^t}{\ln 3} - \frac{2^{3t+1}}{3 \ln 2} - \frac{5}{7} e^{-7t} \quad \frac{2}{\ln 3} - \frac{16}{3 \ln 2} - \frac{5}{7} e^{-7} + \frac{5}{7}$$

$$\int 3^t = \frac{3^t}{\ln 3}$$

$$\int 2^{3t+1} = \int 2^{3t+1} = \frac{2^{3t+1}}{3 \ln 2}$$

$$z = 3t+1 \quad dz/dt = 3$$

$$\frac{dz}{3} = dt$$

$$\int 5e^{-7t} = 5 \int e^{-7t} = -\frac{5}{7} e^{-7t}$$
$$z = -7t \quad dz/dt = -7$$
$$-\frac{dz}{7} = dt$$

Ex 4

$$f(x) = \begin{cases} x^2 + 3^x & x < 2 \\ \sqrt{x} & x \geq 2 \end{cases}$$

$$\int_1^2 x^2 + 3^x = \left[\frac{x^3}{3} + \frac{3^x}{\ln 3} \right]_1^2 = \left(\frac{2^3}{3} + \frac{3^2}{\ln 3} \right) - \left(\frac{1}{3} + \frac{3^1}{\ln 3} \right) = \left(\frac{8}{3} + \frac{9}{\ln 3} \right) - \left(\frac{1}{3} + \frac{3}{\ln 3} \right) = \frac{7}{3} + \frac{6}{\ln 3}$$

$$\int_2^4 \sqrt{x} = \left[\frac{2}{3} \sqrt{x^3} \right]_2^4 = \left(\frac{2}{3} \sqrt{4^3} \right) - \left(\frac{2}{3} \sqrt{2^3} \right) = \left(\frac{2}{3} \cdot 8 \right) - \left(\frac{2}{3} \cdot 2\sqrt{2} \right) = \frac{16}{3} - \frac{4}{3} \sqrt{2} = \frac{4}{3} (4 - \sqrt{2})$$

$$\frac{x^{1/2+1}}{1/2+1} = \frac{x^{3/2}}{3/2} = \frac{2}{3} \sqrt{x^3}$$

$$\int_1^4 f(x) dx = \frac{7}{3} + \frac{6}{\ln 3} + \frac{4}{3} (4 - \sqrt{2})$$

Ex 5

1)

$$f(x) = \sin x \cos^3 x$$

$$\int \sin(x) \cos^3(x) = \int -y^3 dy = -\frac{y^4}{4} = -\frac{\cos^4(x)}{4} + C$$

$$y = \cos(x) \quad \frac{dy}{dx} \cos(x) = -\sin(x)$$

$$dy = -\sin(x) dx$$

$$\frac{dy}{-\sin(x)} = dx$$

$$F(0) = -\frac{\cos^4(0)}{4} + C = 0 \quad -\frac{1}{4} + C = 0 \quad C = \frac{1}{4} \quad F(x) = -\frac{\cos^4(x)}{4} + \frac{1}{4}$$

2)

$$f(x) = \frac{e^{2x+2}}{3e^{2x}-4}$$

$$\int \frac{e^{2x+2}}{3e^{2x}-4} = \int \frac{\cancel{e^{2x}} \cdot e^2}{3e^{2x}-4} \cdot \frac{1}{\cancel{6e^{2x}}} = \int \frac{e^2}{6y} dy = \frac{e^2}{6} \int \frac{1}{y} = \frac{e^2}{6} \ln y = \frac{e^2}{6} \ln (3e^{2x}-4) + C$$

$$y = 3e^{2x} - 4 \quad dy/dx = 6e^{2x}$$

$$dy = 6e^{2x} dx$$

$$\frac{dy}{6e^{2x}} = dx$$

$$F(0) = \frac{e^2}{6} \ln (3e^{2 \cdot 0} - 4) + C = 1 \quad \frac{e^2}{6} \ln (3-4) + C = 1 \quad \frac{e^2}{6} \ln (1) + C = 1 \quad C = 1$$

$$F(x) = \frac{e^2}{6} \ln (3e^{2x} - 4) + 1$$

3)

$$f(x) = \sin(\lg(x))$$

$$\int \sin(\lg(x))$$